

## 63rd ASH Annual Meeting Abstracts

### ORAL ABSTRACTS

#### 722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION

##### Longitudinal Analysis of Gut Bacteriome, Mycobiome, Virome and Metabolome in Allogeneic Stem Cell Transplantation Reveals Susceptibility for Acute Graft-Versus-Host Disease

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**Abstract** The intestinal microbiome directly effects outcomes in patients undergoing allogeneic stem cell transplantation (ASCT) (Peled et al., N Engl J Med, 2020). Recent developments in microbiome research allow us to look beyond bacterial communities at the composition of the intestinal mycobiome and virome. However, only few studies have addressed these fungal (Zhang et al., Nat Commun, 2021; Legoff et al., Nat Med, 2017) and viral communities (Van Der Velden et al., Biol Blood Marrow Transplant, 2013) in the ASCT setting.

Loss of bacterial diversity and domination of single taxa was linked to increased incidence of graft-versus-host-disease (aGVHD) and mortality. However, the functional consequences of intestinal dysbiosis remain poorly understood. Recently, microbial-derived metabolites such as short-chain fatty acids (SCFAs), indoles and secondary bile acids (BA) have been proposed to explain how microbiota exert effects on epithelial and immune responses and shown to protect from GVHD in animal models.

To assemble a complete picture of the phylogenetic and metabolic changes which occur as patients undergo ASCT we designed a multiomics approach and identified patterns in bacteriome, mycobiome, virome and metabolome dynamics in relation to clinical factors and GVHD.

353 stool samples from 80 patients undergoing ASCT were obtained at two different transplantation centers (Munich and Regensburg) at defined timepoints before and after transplantation. Changes in metabolite composition were evaluated via mass spectrometry using a targeted metabolomic profiling approach, accompanied by bacterial 16S rDNA-, fungal 18S-ITS- and deep virome sequencing. Results were linked among each other and with clinical metadata such as GVHD, response to steroids, antimicrobial therapy and mortality.

Longitudinal profiling of ASCT patients revealed a significant loss of bacterial, fungal and viral diversity following ASCT. At both centers, the effect was most pronounced in the early post-transplant period. Patients that developed GI-GVHD had

the lowest diversity scores (**Panel 1**). GVHD patient samples were dominated by *Enterococcus* and *Candida*, and loss of *Caudovirales* species, indicative of deep dysbiosis effecting all intestinal communities.

ASCT had a strong impact on microbiome function and their ability to produce metabolites. We assayed each sample for production of metabolites with reported protective effects: SCFAs (acetate, butyrate and propionate), indoles (indole-3-carboxyaldehyde, ICA) and BAs (deoxycholic and lithocholic acid). When admitted to the transplantation unit (timepoint "conditioning") most patients had rich metabolite expression profiles. In the early post-transplant period, metabolite levels became progressively more depleted. GVHD patients displayed a nearly complete loss of metabolites (**Panel 2a**). Importantly, levels of the metabolite ICA at early timepoints could predict the occurrence of GVHD, leading us to consider prophylactic administration in future studies.

We performed beta diversity analysis to identify clinical factors that had the highest impact on microbial communities. As expected, we noted that administration of broad-spectrum antibiotics had a profound impact of bacteria, fungi and viruses; but also had a significant negative effect on metabolites levels.

Lastly, we report on a case of successful treatment of severe GI-GVHD with fecal microbiota transplant. Response to treatment was characterized by the recipient adopting the FMT donor's rich metabolite expression profile (**Panel 2b**).

To our knowledge, this is the first multicenter study that combines bacteriome, mycobiome, virome and metabolome analyses a longitudinal fashion in matched patient cohorts. Our results are consistent with previous findings that there are major alterations in gut microbiome and metabolite composition after ASCT. We expand upon those findings by characterizing the time course of phylogenetic alterations and their functional consequences. Those alterations play an important role in GvHD pathogenesis and might serve as potential biomarkers to identify patients at high risk for developing lethal GvHD early on. Our results highlight two strategies for targeted microbiome-based interventions: (1) preventing loss of microbial communities by restrictive use of antibiotics and (2) prophylactic administration of metabolite cocktails.

**Disclosures** No relevant conflicts of interest to declare.

<https://doi.org/10.1182/blood-2021-153026>

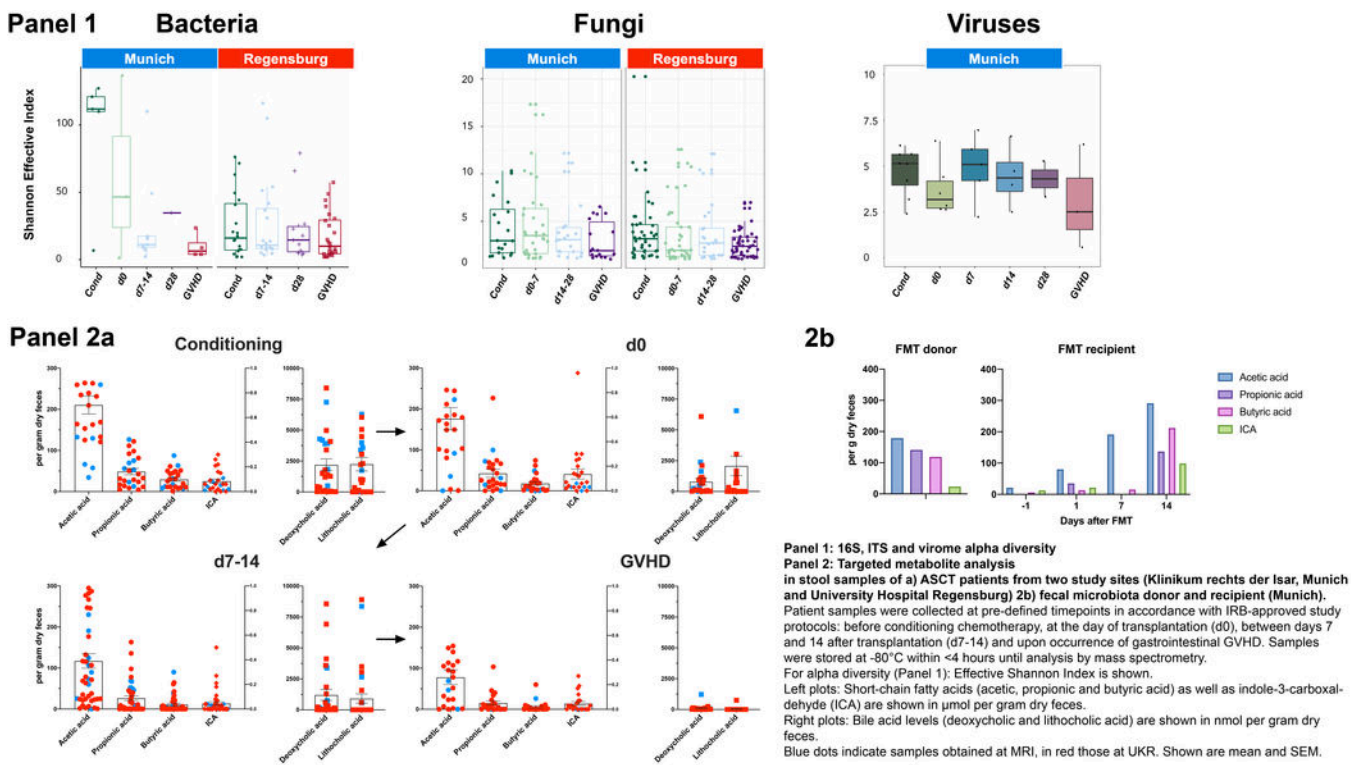


Figure 1